

Telco Customer Churn Analysis

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1. Introduction

Objective

Customer churn directly impacts company revenue and growth. The objective of this project is to analyze customer behavior, identify major churn drivers, and build a predictive model to detect customers at risk of leaving.

Business Problem

A telecom company wants to understand:

- Why customers leave
- Which customers are at high risk
- What actions can reduce churn

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import chi2_contingency
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix, roc_curve, auc
from sklearn.ensemble import RandomForestClassifier
```

```
from google.colab import files
uploaded = files.upload()
```

No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
df = pd.read_excel("Telco_customer_churn.xlsx")
```

2. Dataset Description and Understanding

```
df.head()
```

	CustomerID	Count	Country	State	City	Zip Code	Lat Long	Latitude	Longitude	Gender	...	Contract	Paperless Billing
0	3668-QPYBK	1	United States	California	Los Angeles	90003	33.964131, -118.272783	33.964131	-118.272783	Male	...	Month-to-month	Yes
1	9237-HQITU	1	United States	California	Los Angeles	90005	34.059281, -118.30742	34.059281	-118.307420	Female	...	Month-to-month	Yes
2	9305-CDSKC	1	United States	California	Los Angeles	90006	34.048013, -118.293953	34.048013	-118.293953	Female	...	Month-to-month	Yes
3	7892-POOKP	1	United States	California	Los Angeles	90010	34.062125, -118.315709	34.062125	-118.315709	Female	...	Month-to-month	Yes
4	0280-XJGEX	1	United States	California	Los Angeles	90015	34.039224, -118.266293	34.039224	-118.266293	Male	...	Month-to-month	Yes

5 rows × 33 columns

df.shape

(7043, 33)

Dataset:

- 7043 customer records
- 33 variables

df.columns

```
Index(['CustomerID', 'Count', 'Country', 'State', 'City', 'Zip Code',
      'Lat Long', 'Latitude', 'Longitude', 'Gender', 'Senior Citizen',
      'Partner', 'Dependents', 'Tenure Months', 'Phone Service',
      'Multiple Lines', 'Internet Service', 'Online Security',
      'Online Backup', 'Device Protection', 'Tech Support', 'Streaming TV',
      'Streaming Movies', 'Contract', 'Paperless Billing', 'Payment Method',
      'Monthly Charges', 'Total Charges', 'Churn Label', 'Churn Value',
      'Churn Score', 'CLTV', 'Churn Reason'],
      dtype='object')
```

df.describe()

	Count	Zip Code	Latitude	Longitude	Tenure Months	Monthly Charges	Churn Value	Churn Score	CLTV
count	7043.0	7043.000000	7043.000000	7043.000000	7043.000000	7043.000000	7043.000000	7043.000000	7043.000000
mean	1.0	93521.964646	36.282441	-119.798880	32.371149	64.761692	0.265370	58.699418	4400.295755
std	0.0	1865.794555	2.455723	2.157889	24.559481	30.090047	0.441561	21.525131	1183.057152
min	1.0	90001.000000	32.555828	-124.301372	0.000000	18.250000	0.000000	5.000000	2003.000000
25%	1.0	92102.000000	34.030915	-121.815412	9.000000	35.500000	0.000000	40.000000	3469.000000
50%	1.0	93552.000000	36.391777	-119.730885	29.000000	70.350000	0.000000	61.000000	4527.000000
75%	1.0	95351.000000	38.224869	-118.043237	55.000000	89.850000	1.000000	75.000000	5380.500000

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 33 columns):
#   Column                Non-Null Count  Dtype
---  -
0   CustomerID            7043 non-null   object
1   Count                 7043 non-null   int64
2   Country               7043 non-null   object
3   State                7043 non-null   object
4   City                 7043 non-null   object
5   Zip Code             7043 non-null   int64
6   Lat Long             7043 non-null   object
7   Latitude              7043 non-null   float64
8   Longitude            7043 non-null   float64
9   Gender               7043 non-null   object
10  Senior Citizen        7043 non-null   object
11  Partner              7043 non-null   object
12  Dependents           7043 non-null   object
```

```

13 Tenure Months      7043 non-null int64
14 Phone Service      7043 non-null object
15 Multiple Lines     7043 non-null object
16 Internet Service   7043 non-null object
17 Online Security    7043 non-null object
18 Online Backup      7043 non-null object
19 Device Protection  7043 non-null object
20 Tech Support       7043 non-null object
21 Streaming TV       7043 non-null object
22 Streaming Movies   7043 non-null object
23 Contract           7043 non-null object
24 Paperless Billing   7043 non-null object
25 Payment Method     7043 non-null object
26 Monthly Charges    7043 non-null float64
27 Total Charges      7043 non-null object
28 Churn Label        7043 non-null object
29 Churn Value        7043 non-null int64
30 Churn Score        7043 non-null int64
31 CLTV               7043 non-null int64
32 Churn Reason       1869 non-null object
dtypes: float64(3), int64(6), object(24)
memory usage: 1.8+ MB

```

```
df.isnull().sum()
```

	0
CustomerID	0
Count	0
Country	0
State	0
City	0
Zip Code	0
Lat Long	0
Latitude	0
Longitude	0
Gender	0
Senior Citizen	0
Partner	0
Dependents	0
Tenure Months	0
Phone Service	0
Multiple Lines	0
Internet Service	0
Online Security	0
Online Backup	0
Device Protection	0
Tech Support	0
Streaming TV	0
Streaming Movies	0
Contract	0
Paperless Billing	0
Payment Method	0
Monthly Charges	0
Total Charges	0
Churn Label	0
Churn Value	0
Churn Score	0
CLTV	0
Churn Reason	5174

```
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df["Total Charges"].apply(type).value_counts()
```

	count
Total Charges	
<class 'float'>	6708
<class 'int'>	324
<class 'str'>	11

dtype: int64

Activities Performed

- Examined dataset dimensions
- Reviewed data types
- Checked missing values
- Investigated duplicate records
- Understood business meaning of columns

Observations

- No duplicate customer records found.
- Churn Reason contained missing values because non-churn customers do not have churn reasons.
- Total Charges required datatype correction

3. Data Cleaning & Preparation

```
df["Churn Reason"] = df["Churn Reason"].fillna("No Churn!")
```

Missing values in Churn Reason represent customers who did not churn and were preserved for interpretation.

```
df["Total Charges"] = pd.to_numeric(df["Total Charges"], errors="coerce")
```

```
df["Total Charges"].isnull().sum()
```

```
np.int64(11)
```

```
df[df["Total Charges"].isnull()][  
    "  
    CustomerID",  
    "Tenure Months",  
    "Monthly Charges",  
    "Churn Label"  
]
```

	CustomerID	Tenure	Months	Monthly Charges	Churn Label
2234	4472-LVYGI	0		52.55	No
2438	3115-CZMZD	0		20.25	No
2568	5709-LVOEQ	0		80.85	No
2667	4367-NUYAO	0		25.75	No
2856	1371-DWPAZ	0		56.05	No
4331	7644-OMVMY	0		19.85	No
4687	3213-VVOLG	0		25.35	No
5104	2520-SGTTA	0		20.00	No
5719	2923-ARZLG	0		19.70	No
6772	4075-WKNIU	0		73.35	No
6840	2775-SEFEE	0		61.90	No

```
df["Total Charges"] = df["Total Charges"].fillna(0)
```

Because some rows likely had blank spaces in Total Charges.

A customer may have joined recently has tenure = 0. Therefore no accumulated charges

```
df["Total Charges"].isnull().sum()
```

```
np.int64(0)
```

✓ Cleaning Steps

- Converted Total Charges to numeric format.
- Handled missing values using business logic (churn value).
- Replaced missing Total Charges with 0 for customers with zero tenure.

✓ 4. Exploratory Data Analysis (EDA)

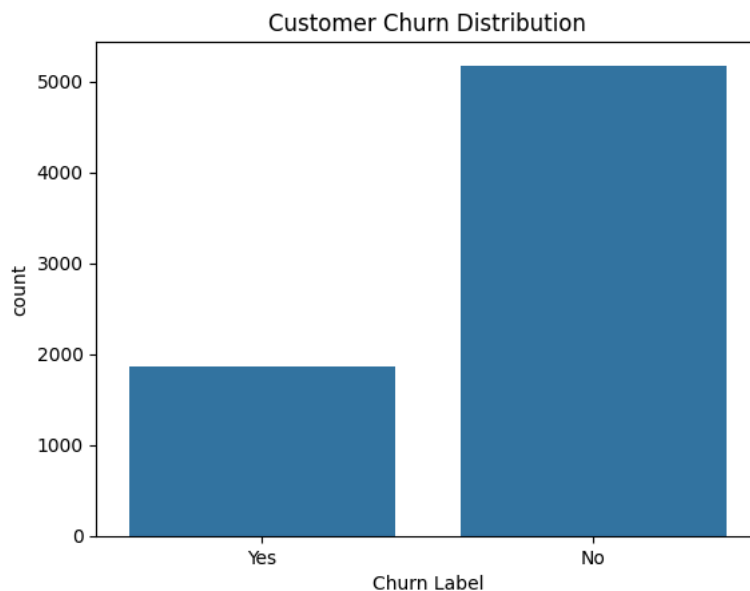
✓ 4.1 Overall Churn Distribution

```
churn = (df["Churn Label"].value_counts(normalize=True)*100)
```

```
print(churn)
```

```
Churn Label
No      73.463013
Yes     26.536987
Name: proportion, dtype: float64
```

```
sns.countplot(x="Churn Label", data=df)
plt.title("Customer Churn Distribution")
plt.show()
```



Insight 1

- Churn Rate = 26.54%
- Approximately 1 in 4 customers left.

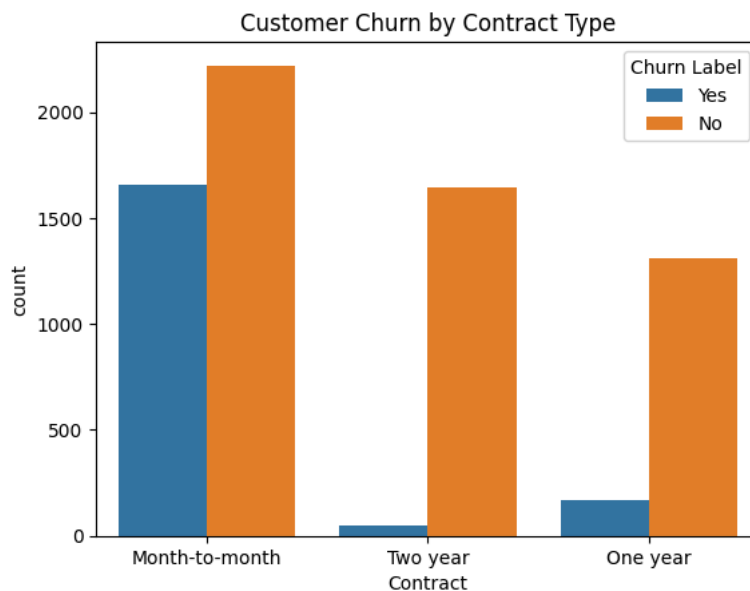
This is not catastrophic, but it is high enough to affect revenue and customer acquisition costs, retention should become a business priority.

4.2 Contract Type Analysis

```
contract_churn = (pd.crosstab(
    df["Contract"],
    df["Churn Label"],
    normalize="index")*100)
contract_churn
```

Churn Label	No	Yes
Contract		
Month-to-month	57.290323	42.709677
One year	88.730482	11.269518
Two year	97.168142	2.831858

```
sns.countplot(x="Contract",hue="Churn Label",data=df)
plt.title("Customer Churn by Contract Type")
plt.show()
```



Insight 2

- Month-to-month customers showed the highest churn (42.7%).
- Two-year contracts showed lowest churn (2.8%).
- Longer commitment appears strongly associated with customer retention.

Encourage month-to-month customers to move into annual plans.

- Offer renewal discounts.
- Create retention campaigns before month 6–12.

4.3 Customer Tenure Analysis

```
df.groupby("Churn Label")["Tenure Months"].mean()
```

Tenure Months	
Churn Label	
No	37.569965
Yes	17.979133

dtype: float64

Insight 3

- Churned customers had average tenure of 18 months.
- Retained customers averaged 38 months.
- First year experience matters a lot.
- Onboarding, support, and service quality may influence retention.

Customer retention risk is highest during the early customer lifecycle.

4.4 Monthly Charges Analysis

```
df.groupby("Churn Label")["Monthly Charges"].mean()
```

Monthly Charges

Churn Label	
No	61.265124
Yes	74.441332

dtype: float64

Insight 4

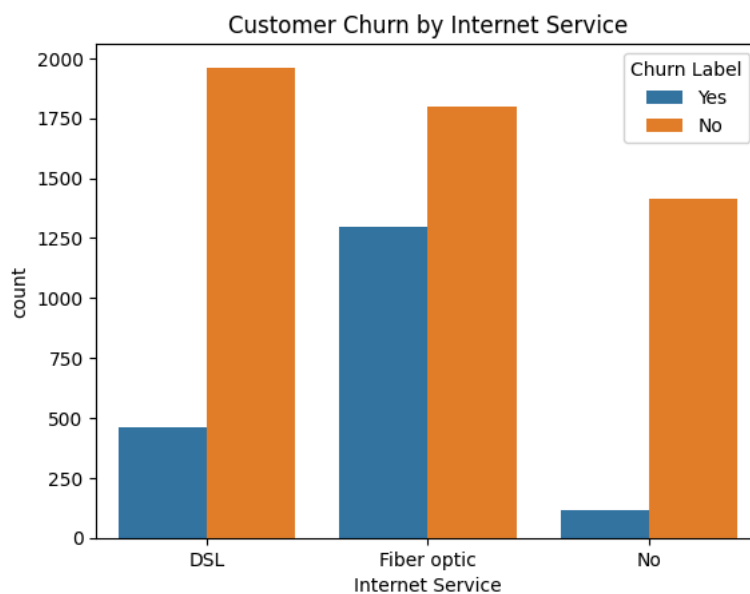
- Churned customers paid higher monthly charges.

4.5 Internet Service Analysis

```
pd.crosstab(
    df["Internet Service"],
    df["Churn Label"],
    normalize="index")*100
```

Churn Label	No	Yes
Internet Service		
DSL	81.040892	18.959108
Fiber optic	58.107235	41.892765
No	92.595020	7.404980

```
sns.countplot(x="Internet Service", hue="Churn Label", data=df)
plt.title("Customer Churn by Internet Service")
plt.show()
```



Insight 5

- Fiber optic customers showed highest churn (41.9%).
- Fiber optic customers are associated with significantly higher churn and should be investigated further.

Since fiber is usually a premium service, this could indicate:

- pricing dissatisfaction
- service quality issues
- competitor pressure
- unmet expectations

4.6 Service Adoption Analysis

```
pd.crosstab(  
    df["Online Security"],  
    df["Churn Label"],  
    normalize="index"  
)*100
```

	Churn Label	No	Yes
Online Security			
	No	58.233276	41.766724
No internet service	92.595020	7.404980	
Yes	85.388806	14.611194	

Customers without security service churn nearly 3× more.

- Bundle security with plans
- Offer free trials

```
pd.crosstab(  
    df["Tech Support"],  
    df["Churn Label"],  
    normalize="index"  
)*100
```

	Churn Label	No	Yes
Tech Support			
	No	58.364526	41.635474
No internet service	92.595020	7.404980	
Yes	84.833659	15.166341	

Customers with technical support remain substantially longer.

Support may act as a retention mechanism, not only a service expense.

Insight 6

Customers with:

- Online Security
- Tech Support

showed significantly lower churn.

4.7 Payment Method Analysis

```
pd.crosstab(  
    df["Payment Method"],  
    df["Churn Label"],  
    normalize="index"  
)*100
```

	Churn Label	No	Yes
Payment Method			
Bank transfer (automatic)	83.290155	16.709845	
Credit card (automatic)	84.756899	15.243101	
Electronic check	54.714588	45.285412	
Mailed check	80.893300	19.106700	

Insight 7

- Electronic check users showed highest churn (45.3%).

Electronic check customers churn almost 3× more than automatic payment users.

Possible explanations:

- Lower engagement
- Less convenience
- Short-term customer behavior

Migration towards auto-payment should be encouraged.

Now, we have something meaningful to conclude

✓ High-Risk Customer Profile

- Month-to-month contract
 - Short tenure
 - High monthly charges
 - Fiber internet
 - No online security
 - No tech support
 - Payment method
-

✓ 5. Statistical Validation

✓ Chi-Square Test

- H_0 : Contract type and churn are independent.
- H_1 : Contract type and churn are associated.

```
table = pd.crosstab(
    df["Contract"],
    df["Churn Label"]
)

chi2, p, dof, expected = chi2_contingency(table)
```

Chi-Square Statistic (χ^2)

- Measures how different the observed values are from the expected values. Higher value : stronger evidence that variables are related.

P-value (p)

- Tells whether the relationship found is statistically significant or may have happened by chance.

Degrees of Freedom (dof)

- Number of independent values used in the calculation.
- Formula: $(\text{rows} - 1) \times (\text{columns} - 1)$.

Basically a map to let the code knows it can't go beyond it, every event would occur inside it.

✓ Expected Frequencies (expected)

- The values we would expect if there were no relationship between the variables (null hypothesis is true).

```
print("p-value:", p)
```

```
p-value: 5.863038300673391e-258
```

✓ P-value Decision

- $p < 0.05$ Reject Null Hypothesis → Significant relationship exists
- $p \geq 0.05$ Fail to Reject Null Hypothesis → No significant relationship

✓ Conclusion

Contract type showed statistically significant association with churn.

The Chi-Square test was performed to determine whether an association exists between the selected categorical variables. The obtained p-value (5.863e-258) is significantly lower than 0.05, indicating strong statistical evidence against the null hypothesis. Therefore, it can be concluded that a significant relationship exists between the variables.

✓ 6. Predictive Modeling

```
drop_cols = [  
    "CustomerID",  
    "Count",  
    "Country",  
    "State",  
    "City",  
    "Zip Code",  
    "Lat Long",  
    "Latitude",  
    "Longitude",  
    "Churn Label",  
    "Churn Score",  
    "Churn Reason"  
]  
  
model_df = df.drop(columns=drop_cols)
```

```
model_df = pd.get_dummies(  
    model_df,  
    drop_first=True  
)
```

```
model_df.shape
```

```
(7043, 32)
```

```
X = model_df.drop(  
    "Churn Value",  
    axis=1)  
  
y = model_df["Churn Value"]  
  
X_train, X_test, y_train, y_test = (  
    train_test_split(  
        X,  
        y,  
        test_size=0.2,  
        random_state=42,  
        stratify=y))
```

✓ Model 1 — Logistic Regression

```
model = LogisticRegression(  
    max_iter=5000  
)  
  
model.fit(  
    X_train,  
    y_train
```

```
X_train,
y_train
)
```

/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
```

```
    LogisticRegression
```

```
LogisticRegression(max_iter=5000)
```

```
pred = model.predict(X_test)
```

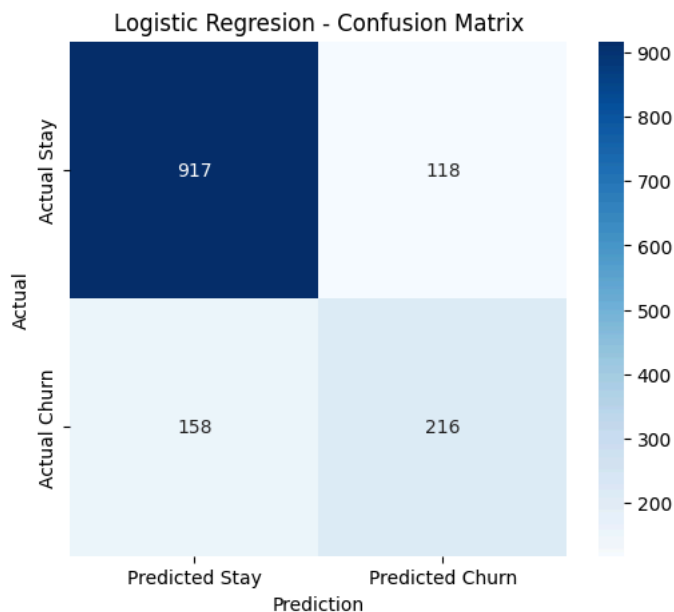
```
print(classification_report(y_test,pred))
```

	precision	recall	f1-score	support
0	0.85	0.89	0.87	1035
1	0.65	0.58	0.61	374
accuracy			0.80	1409
macro avg	0.75	0.73	0.74	1409
weighted avg	0.80	0.80	0.80	1409

The model demonstrated strong capability in identifying customers likely to churn while remaining interpretable.

```
cm = confusion_matrix(y_test,pred)
plt.figure(figsize=(6,5))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["Predicted Stay", "Predicted Churn"],
    yticklabels=["Actual Stay", "Actual Churn"]
)

plt.title("Logistic Regresion - Confusion Matrix")
plt.xlabel("Prediction")
plt.ylabel("Actual")
plt.show()
```



- True Negatives (917)

These customers stayed and the model correctly predicted they would stay.

- True Positives (216)

These customers churned and the model successfully detected them.

This is valuable because these are customers the business could target.

- False Positives (118)

Model predicted churn but customer actually stayed.

Some retention offers may be wasted.

- False Negatives (158)

These are important. Customer churned but model predicted stay.

Company misses opportunity to intervene.

✓ ROC Curve

After evaluating the model using accuracy, precision, recall, and F1-score, the ROC Curve is used to measure how well the model distinguishes between churn and non-churn customers across different classification thresholds.

The ROC curve provides a visual representation of model performance, while the AUC (Area Under Curve) score summarizes the model's ability to correctly separate the two classes.

A higher AUC value indicates better classification performance.

```
probs = model.predict_proba(
    X_test
)[: ,1]

fpr,tpr,_ = roc_curve(
    y_test,
    probs
)

roc_auc = auc(
    fpr,
    tpr
)

plt.figure(figsize=(7,5))

plt.plot(
    fpr,
    tpr,
    label=f"AUC = {roc_auc:.3f}"
)

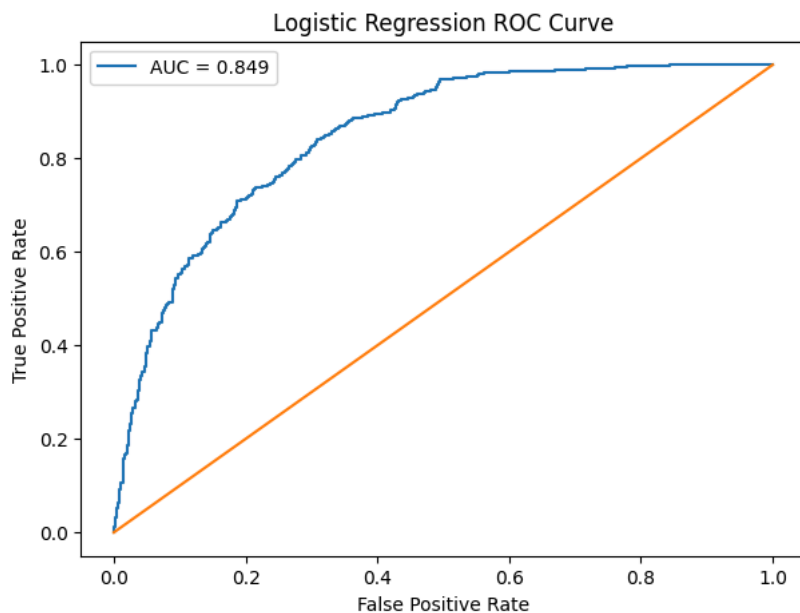
plt.plot(
    [0,1],
    [0,1]
)

plt.xlabel(
    "False Positive Rate"
)

plt.ylabel(
    "True Positive Rate"
)

plt.title(
    "Logistic Regression ROC Curve"
)

plt.legend()
plt.show()
print(roc_auc)
```



0.8493296132682322

The Logistic Regression model achieved an ROC-AUC score of 0.849, indicating very good performance in distinguishing customers who are likely to churn from those who are likely to stay. This result shows that the model performs significantly better than random guessing and provides reliable support for customer churn prediction.

Model 2 — Random Forest

```
rf = RandomForestClassifier(  
    n_estimators=300,  
    random_state=42  
)  
  
rf.fit(X_train, y_train)  
pred_rf = rf.predict(X_test)
```

```
print(classification_report(  
    y_test,  
    pred_rf  
))
```

	precision	recall	f1-score	support
0	0.84	0.89	0.86	1035
1	0.64	0.52	0.57	374
accuracy			0.79	1409
macro avg	0.74	0.70	0.72	1409
weighted avg	0.78	0.79	0.79	1409

Random Forest did not outperform Logistic Regression

The model achieved 79% accuracy and identified important factors such as Total Charges, Tenure Months, and Monthly Charges. Although Random Forest performed well, Logistic Regression provided better recall and stronger interpretability. Therefore, Logistic Regression was selected as the final model for this project.

Why Predictive Modeling Was Performed

- After identifying patterns through exploratory data analysis, predictive modeling was performed to estimate whether a customer is likely to churn based on customer attributes and service usage. Since the target variable (Churn Value) contains two categories (0 = Stay, 1 = Churn), this became a classification problem.
- Logistic Regression was selected as the first model because it is simple, interpretable, and commonly used for binary classification problems. It helps understand which factors are associated with higher churn probability.

- Random Forest was selected because it is a powerful classification algorithm that works well with structured business data and can identify complex relationships between customer features. It was chosen to compare performance against Logistic Regression and check whether a more advanced model could improve churn prediction. Random Forest also provides feature importance, which helps explain business insights.
 - Other algorithms were not used because the goal of this project was to build an accurate and understandable business solution rather than maximize complexity. Logistic Regression and Random Forest provided a good balance between performance and interpretability.
-

7. Feature Importance

Feature Importance

- it identify which customer attributes has the greatest influence on churn prediction. Random Forest assigns importance scores to each feature based on how much it contributes to decision-making. Higher importance values indicate stronger influence on predicting whether a customer is likely to churn.

```
importance=(  
pd.Series(  
    rf.feature_importances_,  
    index=X.columns  
)  
)  
.sort_values(ascending=False)  
)  
print(importance.head(10))
```

Total Charges	0.158992
Tenure Months	0.146733
Monthly Charges	0.134460
CLTV	0.124381
Internet Service_Fiber optic	0.041953